

BAULKHAM HILLS HIGH SCHOOL MARKING COVER SHEET.



YEAR 12 Trial EXTENSION 2 2011

STUDENT NUMBER: _____

TEACHER NAME: _____

QUESTION	MARK
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	/ 120
PERCENTAGE	%



BAULKHAM HILLS HIGH SCHOOL

2011

**TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work

Total marks – 120

- Attempt Questions 1 – 8
- All questions are of equal value
- Start a separate piece of paper for each question.
- Put your student number and the question number at the top of each sheet.

Question 1 (15 marks) - Start on a new page		Marks
a)	Find the following indefinite integrals	
	(i) $\int \cos^3 x \, dx$	2
	(ii) $\int \frac{x-2}{x^2+1} \, dx$	2
	(iii) $\int x \sin 2x \, dx$	2
b)	Evaluate $\int_0^1 \frac{dx}{\sqrt{3-2x-x^2}}$	3
c)	Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{1+\sin x+\cos x}$	3
d)	Find $\int \frac{2x \, dx}{x^3-2x^2+9x-18}$	3

Question 2 (15 marks) - Start on a new page		
a)	(i) Express $z_1 = \frac{7+4i}{3-2i}$ in the form $a+ib$ where a, b are real (ii) On an Argand diagram sketch the locus of the point representing the complex number z such that $z - z_1 = \sqrt{5}$ (iii) Prove that the locus passes through the origin and find the greatest value of z	2 1 2
b)	Let $z = 2 + 3i$ and $w = 1 + i$ Find zw and $\frac{1}{w}$ in the form $x + iy$	2
c)	(i) Express $(1 - \sqrt{3}i)$ in modulus argument form (ii) Hence write $(1 - \sqrt{3}i)^{10}$ in the form $x + iy$	2 2
d)	The complex number $z = x + iy$ when x and y are real, is such that $z - i = \text{Im}(z)$ (i) Show that the locus of point P representing z has Cartesian equation $y = \frac{1}{2}(x^2 + 1)$ and sketch the locus (ii) By finding the gradients of the tangents to this curve which pass through the origin, find the set of possible values of $\arg z$ ($-\pi < \arg z \leq \pi$)	2 2

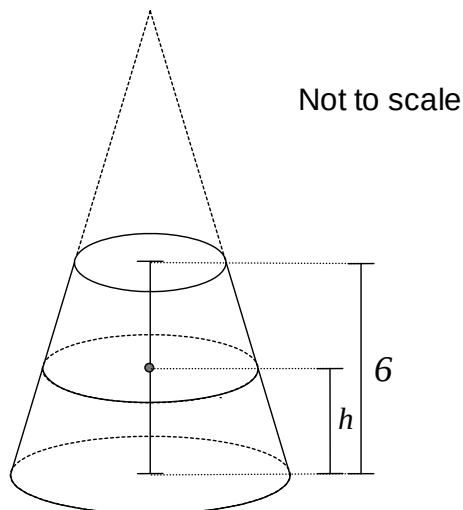
Question 3 (15 marks) - Start a new page		Marks
a)	Sketch the graph $y = (2x + 1)(x + 1)$ clearly showing all intercepts on the co-ordinate axes and the co-ordinates of any turning points.	2
b)	Use the graph of part (a) to sketch the graphs below, showing clearly the intercepts on the co-ordinate axes, the co-ordinates of any turning points and the equation of any asymptotes. (i) $y = \log_e[(2x + 1)(x + 1)]$ (ii) $y = \frac{1}{(2x + 1)(x + 1)}$	2 2
c)	The region bounded by the curve $y = \frac{1}{(2x + 1)(x + 1)}$ the co-ordinate axes and the line $x = 4$ is rotated through one complete revolution about the y axis. (i) Use the method of cylindrical shells to express the volume of the resulting solid of revolution as an integral . (ii) Evaluate the integral in part (i).	2 4
d)	When $P(x) = x^4 + ax^3 + b$ is divided by $x^2 + 4$, the remainder is $-x + 13$. Find the values of a and b.	3

Question 4 (15 marks) - Start a new page

- a) A hyperbola has Cartesian equation $3x^2 - y^2 = 12$
- find
- (i) its eccentricity 1
 - (ii) the co-ordinates of its foci 1
 - (iii) the equations of the directrices 1
 - (iv) the equations of the asymptotes 1

hence sketch the hyperbola indicating all the features of your diagram. 1

- b) A right elliptical cone has its top cut off through a plane parallel to its elliptical base. The remaining solid has an ellipse as its base



The remaining solid has the ellipse $\frac{x^2}{9} + \frac{4y^2}{9} = 1$ as its base, and another ellipse $x^2 + 4y^2 = 1$ as its top.

The height of the solid is 6 units.

- (i) Given that the area of an ellipse with equation $\frac{x^2}{9} + \frac{4y^2}{9} = 1$ is πab , show that the area of the ellipse at height h units above the base is $A = \frac{\pi(h-9)^2}{18}$ 3
- (ii) Hence find the volume of the solid. 3

- c) A plane curve is defined implicitly by $x^2 + 2xy + y^5 = 4$
This curve has a horizontal tangent at $P(x, y)$ show that $x = \alpha$ is a root of the equation $x^5 + x^2 + 4 = 0$ 4

Question 5 (15 marks) - Start a new page	Marks
a) The equation $x^3 + px - 1 = 0$ has 3 non zero roots α, β, γ (i) Find the values of $\alpha^2 + \beta^2 + \gamma^2$ and $\alpha^4 + \beta^4 + \gamma^4$ in terms of p and show that p must be negative. 4 (ii) Find the monic equation with coefficients in terms of p where roots are $\frac{\alpha}{\beta\gamma}, \frac{\beta}{\alpha\gamma}, \frac{\gamma}{\alpha\beta}$ 2	
b) A chord PQ of the rectangular hyperbola $xy = 9$ meets the asymptotes at L and M as shown i) Show that the equation of the chord PQ is $pqy + x = 3(p + q)$ 1 ii) Find the co-ordinates of N the midpoint of PQ 2 iii) Show that $PL = MQ$ 2 iv) If the chord PQ is a tangent to the parabola $y^2 = 3x$ find the locus of N 2	
c) Solve in terms of a $a^x = e^{2x-1}$ where $a > 0, a \neq \frac{1}{e^2}$ 2	

Question 6 (15 marks) - Start a new page		
a)	Solve the equation $x^4 - 6x^3 + 9x^2 + 6x - 20 = 0$ given $(2 + i)$ is one of its zeroes.	3
b)	$P(a \cos \theta, b \sin \theta)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{where } a > b > 0.$ The tangent and normal at point P cut y-axis at A and B respectively, and S is a focus of the ellipse i) Show that $\angle ASB = 90^\circ$ ii) Hence show that A, P, S and B are concyclic and state the coordinates of the centre of the circle through A, P, S and B.	2 3
c)	Prove that $\int_0^a f(x) \, dx = \int_0^a f(a - x) \, dx$ Hence evaluate $\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} \, dx$	2 3
d)	Draw a neat sketch of $y = \frac{1}{\sin^{-1} x}$	2

Question 7 (15 marks) - Start a new page		Marks
a)	Given that $1, \omega$ and ω^2 are the three cube roots of unity i) Find the value of $(1 + 2\omega + 3\omega^2)(1 + 2\omega^2 + 3\omega)$ ii) If the equations $x^3 - 1 = 0$ and $px^5 + qx + r = 0$ have a common root, evaluate $(p + q + r)(p\omega^5 + q\omega + r)(p\omega^{10} + q\omega^2 + r)$	3 3
b)	i) Show that $(1 - \sqrt{x})^{n-1} \cdot \sqrt{x} = (1 - \sqrt{x})^{n-1} - (1 - \sqrt{x})^n$ ii) If $I_n = \int_0^1 (1 - \sqrt{x})^n dx$ for $n \geq 0$ Show that $I_n = \frac{n}{n+2} I_{n-1}$ for $n \geq 1$ And hence evaluate I_{100}	1 3
c)	Prove that the volume, V, the area of the curved surface, S, and the radius of the base, r, of a right circular cone are connected by the equation $9V^2 = r^2(S^2 - \pi^2 r^4)$ Show that the maximum volume for a given curved surface area S, is $\frac{2^{\frac{1}{2}} S^{\frac{3}{2}}}{\pi^{\frac{1}{2}} 3^{\frac{7}{4}}}$	2 3

Question 8 (15 marks) - Start a new page

a) Prove by mathematical induction that $7^n + 3n(7^n) - 1$ is divisible by 9.

3

b) i) Write the general solution of

$$\tan 4\theta = 1$$

1

ii) Use De Moivre's Theorem to find $\cos 4\theta$ and $\sin 4\theta$ in terms of $\cos \theta$ and $\sin \theta$ and hence determine the result

4

$$\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$$

iii) Find the roots of

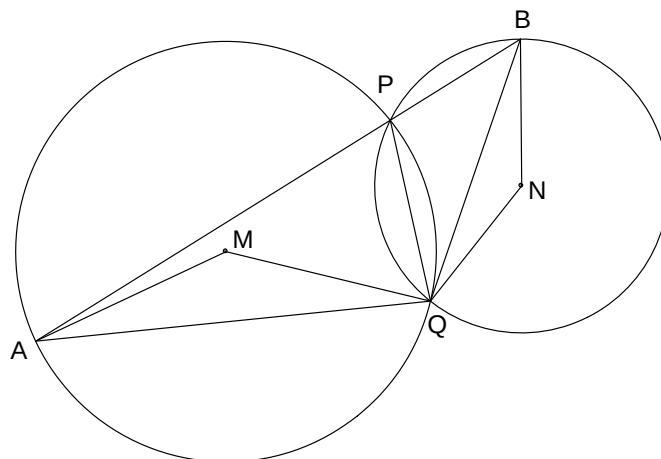
$$x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$$

in the form $x = \tan \theta$ and hence prove that

3

$$\tan^2 \frac{\pi}{16} + \tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{7\pi}{16} = 28$$

c)



4

In the given diagram, two circles whose centres are M and N intersect in P and Q . A line drawn through P meets the two circles in A and B . Prove that $\angle MAQ = \angle NBQ$.

End of Exam

$$\begin{aligned}
 \text{Q. 1)} \quad & \int \cos^3 x \, dx \\
 \text{①} \quad & = \int \cos x (1 - \sin^2 x) \, dx \\
 & = \int \cos x \, dx - \int \sin^2 x \cos x \, dx \\
 & = \sin x - \frac{\sin^3 x}{3} + C
 \end{aligned}$$

$$\begin{aligned}
 \text{II)} \quad & \int \frac{x-2}{x^2+1} \, dx \\
 & = \int \frac{x}{x^2+1} \, dx - \int \frac{2}{x^2+1} \, dx \\
 & = \frac{1}{2} \ln(x^2+1) - 2 \tan^{-1} x + C
 \end{aligned}$$

$$\begin{aligned}
 \text{III)} \quad & \int x \sin 2x \, dx \\
 \text{let } u &= x \quad v' = \sin 2x \\
 u' &= 1 \quad v = -\frac{1}{2} \cos 2x \\
 \therefore I &= -\frac{1}{2} x \cos 2x + \frac{1}{2} \int \cos 2x \, dx \\
 &= -\frac{1}{2} x \cos 2x + \frac{1}{4} \sin 2x + C
 \end{aligned}$$

$$\begin{aligned}
 \text{b)} \quad & \int_0^1 \frac{dx}{\sqrt{-x^2+2x+3}} \\
 &= \int_0^1 \frac{dx}{\sqrt{-(x^2-2x+1)+4}} \\
 &= \int_0^1 \frac{dx}{\sqrt{4-(x-1)^2}} \\
 &= \left[\sin^{-1} \left(\frac{x-1}{2} \right) \right]_0^1 \\
 &= \sin^{-1} 1 - \sin^{-1} \frac{1}{2} \\
 &= \frac{\pi}{2} - \frac{\pi}{6} \\
 &= \frac{\pi}{3}
 \end{aligned}$$

$$\begin{aligned}
 \text{c)} \quad & \int_0^{\pi/2} \frac{dx}{1+\cos x + \sin x} \quad \text{①} \\
 \text{let } t &= \tan \frac{x}{2} \quad x = \frac{\pi}{2} \quad t = 1 \\
 & \quad \quad \quad x = 0 \quad t = 0 \\
 I &= \int_0^1 \left(\frac{1}{1 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}} \right) \cdot \frac{2dt}{1+t^2} \\
 &= \int_0^1 \frac{2dt}{1+t^2+1-1+t^2} \\
 &= \int_0^1 \frac{dt}{1+t} \\
 &= \left[\ln(1+t) \right]_0^1 \\
 &= \ln 2 - \ln 1 \\
 &= \ln 2
 \end{aligned}$$

$$\begin{aligned}
 \text{d)} \quad & \int \frac{2x \, dx}{x^3-2x^2+9x-18} \\
 &= \int \frac{2x \, dx}{(x^2+9)(x-2)}
 \end{aligned}$$

$$\begin{aligned}
 \text{let } \frac{2x}{(x-2)(x^2+9)} &\equiv \frac{A}{x-2} + \frac{Bx+C}{x^2+9} \\
 A(x^2+9) + (Bx+C)(x-2) &\equiv 2x \\
 x=2 \rightarrow 13A &= 4 \quad \therefore A = \frac{4}{13} \\
 \text{Coeff of } x^2 \quad 9A - 2C &= 0 \\
 2C &= 9 \times \frac{4}{13} \quad C = \frac{18}{13} \\
 A+B &= 0 \quad B = -\frac{4}{13} \\
 \therefore I &= \frac{1}{13} \left[\int \frac{4}{x-2} \, dx + \int \frac{18-4x}{x^2+9} \, dx \right] \\
 &= \frac{1}{13} \left[4 \ln(x-2) + \frac{18}{3} \tan^{-1} \frac{x}{3} - 2 \ln(x^2+9) \right] + C \\
 &= \frac{4}{13} \ln(x-2) + \frac{6}{13} \tan^{-1} \frac{x}{3} - \frac{2}{13} \ln(x^2+9) + C
 \end{aligned}$$

$$a) \quad z = 2+3i \quad w = (1+i)$$

$$zw = (2+3i)(1+i)$$

$$= -1+5i$$

$$\frac{1}{w} = \frac{1}{1+i} \times \frac{1-i}{1-i}$$

$$= \frac{1-i}{2}$$

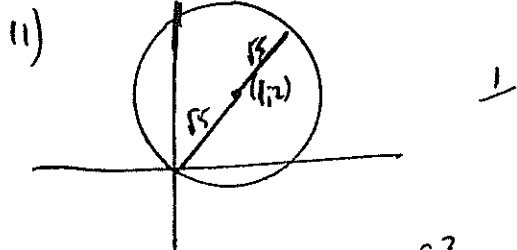
$$= \frac{1}{2} - \frac{1}{2}i$$

$$b) \quad z_1 = \frac{7+4i}{3-2i} \times \frac{3+2i}{3+2i}$$

$$= \frac{21+14i+12i-8}{9+4}$$

$$= \frac{13+26i}{13}$$

$$= 1+2i$$



$$\text{eqn of circle is } (x-1)^2 + (y-2)^2 = 5^2$$

$$\therefore x^2 - 2x + 1 + y^2 - 4y + 4 = 5$$

$$x^2 + y^2 - 2x - 4y = 0$$

$$x=0, y=0 \text{ satisfies the eqn.}$$

$$\text{num } |z| = \text{distance of circle}$$

$$1 \text{ or } 2\sqrt{5}$$

$$c) \quad i) \quad (1-\sqrt{3}i)$$

$$= 2\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)$$

$$= 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

$$\text{or } 2e^{i\frac{\pi}{3}}$$

$$ii) \quad (1-\sqrt{3}i)^8$$

$$= 2^8 e^{i\frac{8\pi}{3}}$$

$$= 2^8 e^{i\frac{2\pi}{3}}$$

$$= 2^8 \left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$$

$$= 2^8 \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)$$

$$= -128 - 128\sqrt{3}i$$

$$d) \quad i) \quad |3-i| = \rho_m z$$

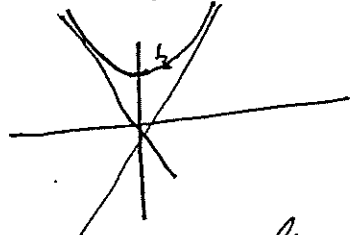
$$(x+iy-i) = y$$

$$x^2 + (y-1)^2 = y^2$$

$$x^2 + y^2 - 2y + 1 = y^2$$

$$x^2 - 2y + 1 = 0$$

$$y = \frac{1}{2}(x^2 + 1)$$



$$ii) \quad \text{equation of tangent line } 0 \text{ is}$$

$$y = mx$$

$$mx = \frac{1}{2}(x^2 + 1)$$

$$x^2 - 2mx + 1 = 0$$

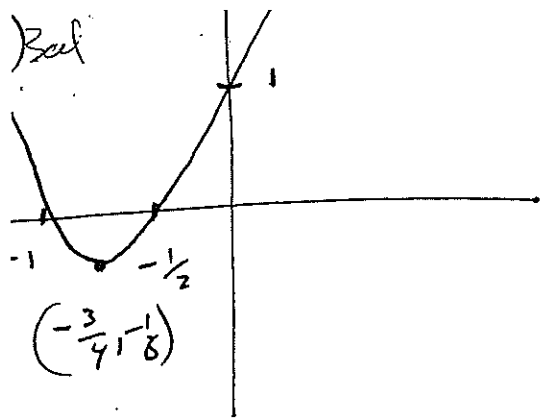
$$\text{only one root}$$

$$\therefore \Delta = 0$$

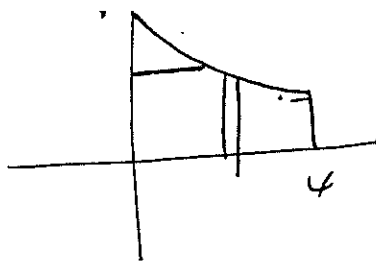
$$4m^2 - 4 = 0$$

$$m = \pm 1$$

$$\therefore \frac{\pi}{4} \leq \arg z \leq \frac{3\pi}{4}$$



2/



3

the thickness of shell be δx .

$$\delta V = 2\pi x \cdot y \cdot \delta x$$

$$= 2\pi x \frac{dx}{(2x+1)(x+1)}$$

$$V = \sum_{x=0}^{x=4} \delta V$$

$$= 2\pi \int_0^4 \frac{x dx}{(2x+1)(x+1)}$$

$$\text{let } \frac{x}{(2x+1)(x+1)} = \frac{a}{2x+1} + \frac{b}{x+1}$$

$$a = -1 \quad b = 1$$

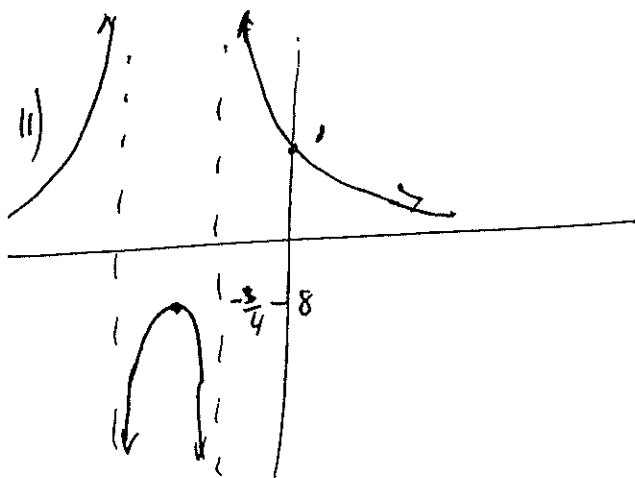
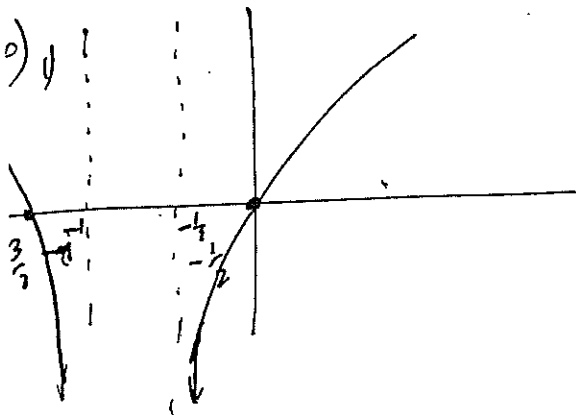
$$\therefore V = 2\pi \int_0^4 \left(\frac{1}{x+1} - \frac{1}{2x+1} \right) dx$$

$$= 2\pi \left[\ln(x+1) - \frac{1}{2} \ln(2x+1) \right]_0^4$$

$$= 2\pi \left\{ \ln 5 - \ln 1 - \frac{1}{2} \ln 9 + \frac{1}{2} \ln 1 \right\}$$

$$= 2\pi \left\{ \ln 5 - \frac{1}{2} \ln 9 \right\}$$

$$= 2\pi \ln \frac{5}{3}$$



1) $x^4 + ax^3 + b = (x^2 + 4) Q(x) + -x + 13$

let $x=2i$ $16 - 8ai + b = -2i + 13$

$$\therefore a = \frac{1}{4}$$

$$b = -3$$

④

4) $3x^2 - y^2 = 12$

$\frac{x^2}{4} - \frac{y^2}{12} = 1$

$a=2 \quad b=2\sqrt{3}$

$b^2 = a^2(e^2 - 1)$

$12 = 4(e^2 - 1)$

$e^2 - 1 = 3$

$e = 2 \quad \text{--- (1)}$

5) $(4, 0) \quad S'(-4, 0) \quad \text{--- (1)}$

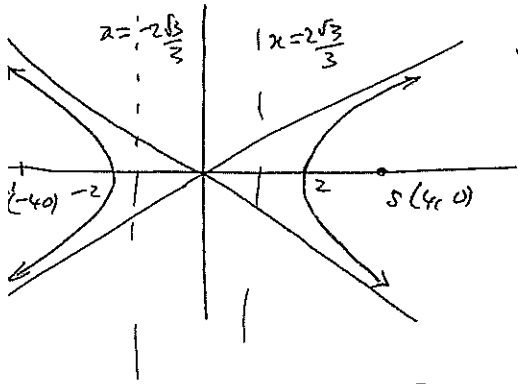
~~$y = \pm \sqrt{3}x$~~

rectrices

~~$y = x = \pm \frac{a}{e}$~~

$x = \pm \frac{2}{\sqrt{3}}$

or $x = \pm \frac{2\sqrt{3}}{3}$



$y = \pm \sqrt{3}x$ (asymptotes)

c) $x^2 + 2xy + y^5 = 4$

$2x + 2y + 2xy' + 5y^4y' = 0$

$y'(2x + 5y^4) = -(2x + 2y)$

$y' = \frac{-2(x+y)}{2x+5y^4}$

$= 0$ when ~~$x = -y$~~
 $y = -x$

subst $y = -x$ in eqn

$\therefore x^2 + 2x(-x) + (-x)^5 = 4$

$x^2 - 2x^2 - x^5 = 4$

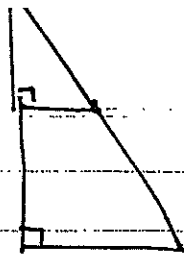
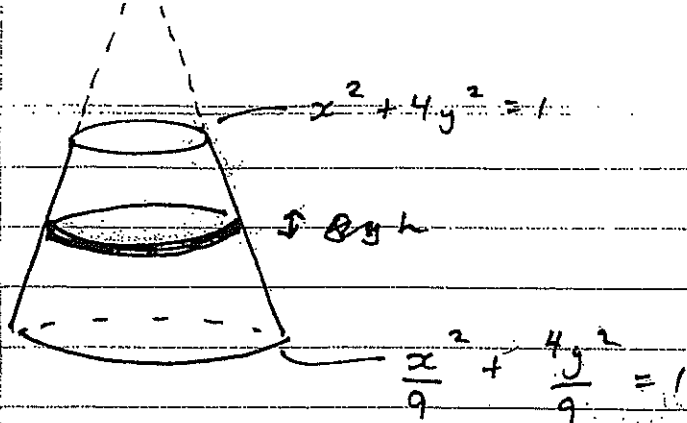
$-x^2 - x^5 = 4$

$x^5 + x^2 + 4 = 0$

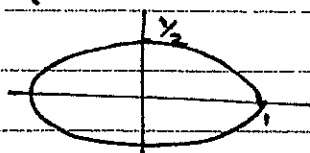
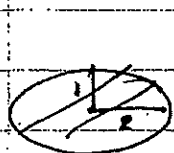
$\frac{b^2 + b^2x^2 - a^2x^2 + a^2}{2}$

$\frac{b^2}{9} + \left(\frac{a^2}{b^2}x^2\right) + \frac{a^2x^2}{b^2}$

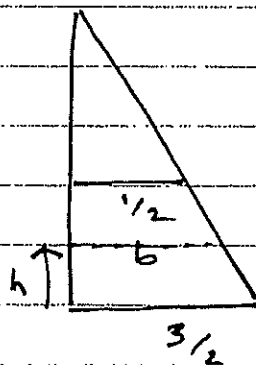
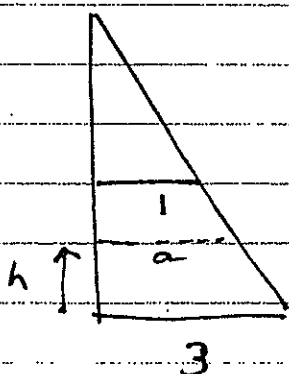
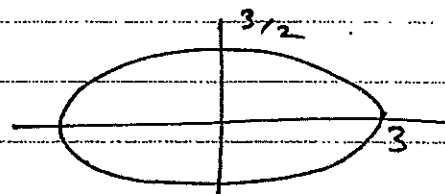
46)



1) Show that the volume of the cylindrical disc at height h is



$$x^2 + 4y^2 = 1$$



$$a = mh + b$$

$$\text{If } h = 0, a = 3$$

$$3 = 0 + b \therefore b = 3$$

$$\text{If } h = 6, a = 1$$

$$1 = 6m + 3$$

$$-2 = 6m$$

$$m = -1/3$$

$$\therefore a = -\frac{1}{3}h + 3$$

$$b = mh + c$$

$$\text{If } h = 0, b = 3/2$$

$$\frac{3}{2} = 0 + c$$

$$c = 3/2$$

$$\text{If } h = 6, b = 1/2$$

$$\frac{1}{2} = 6m + \frac{3}{2}$$

$$-1 = 6m$$

$$m = -1/6$$

$$\therefore b = -\frac{1}{6}h + \frac{3}{2}$$

$$\therefore \text{Area of ellipse at height } h = \pi \left(-\frac{1}{3}h + 3 \right) \left(-\frac{1}{6}h + \frac{3}{2} \right)$$

$$\therefore \Delta V = \pi$$

$$= \pi \cdot -\frac{1}{3}(h-9) \cdot -\frac{1}{6}(h-9)$$

$$= \frac{1}{18} \pi (h-9)^2$$

$$\times \square = \frac{3}{2}$$

$$\square = \frac{3}{2} \times 6$$

$$\therefore \Delta V = \frac{1}{18} \pi \cdot (h-9)^2 \cdot \Delta h$$

$$V = \frac{\pi}{18} \int_0^6 (h-9)^2 \cdot dh$$

$$= \frac{\pi}{18} \cdot \left[\frac{(h-9)^3}{3} \right]_0^6$$

$$= \frac{\pi}{18} \left(\frac{(-3)^3}{3} - \frac{(-9)^3}{3} \right) \quad \frac{729}{3}$$

$$= \frac{\pi}{18} (-9 + 243)$$

$$= \frac{234\pi}{18}$$

$$= 13\pi$$

(i) ~~Given that the area of an ellipse is πab units², Show that the area~~

③

$$a) 1) \Sigma x^2 = (\Sigma x)^2 - 2 \Sigma xB$$

$$\Sigma x = 0 \quad \Sigma xB = p \quad \perp$$

Left $\lambda = 0 \quad y = \frac{3(p+q)}{pq}$

$$\therefore \Sigma x^2 = 0 - 2p = -2p \quad \perp$$

Right $y = 0 \quad x = 3(p+q)$

$$x^3 = -px + 1$$

$$x^4 = -px^2 + x$$

1. MP of LM is $\frac{3(p+q)}{pq} \rightarrow 3p$

$$2y = -px^2 + x \quad \perp$$

$$\therefore \Sigma x^4 = -p \Sigma x^2 + \Sigma x$$

$$= -p \cdot -2p + 0$$

$$= 2p^2 \quad \perp$$

$$\frac{3(p+q)}{2}, \frac{3(p+q)}{2pq}$$

$$\Sigma x^2 > 0 \text{ if not one root is zero}$$

$$\therefore -2p > 0 \therefore p < 0$$

" MP of LM is same as MP of QP $\perp$

$$\therefore LP = MQ$$

roots are $\frac{x^2}{LB}, \frac{B^2}{LB}, \frac{y^2}{LB}$

$$pqy + x = 3(p+q) \quad \text{--- (1)}$$

$$y^2 = 3x \quad \text{--- (2)}$$

$$xBy = 1$$

$$\therefore \text{not all } x^2, B^2, y^2 \quad \perp$$

sub $x = \frac{y^2}{3} \Rightarrow (1)$

$$\text{at } y = x^2$$

$$\therefore x = \sqrt{y}$$

$$\therefore y^{\frac{3}{2}} + p y^{\frac{1}{2}} - 1 = 0$$

$$\therefore pqy + \frac{y^2}{3} = 3(p+q)$$

$$y^{\frac{1}{2}}(y+p) = 1$$

$$\therefore y^2 + 3pqy - 9(p+q) = 0$$

$$y(y^2 + 2px + p^2) = 0 \quad \perp$$

there is only one soln $\perp$

$$\therefore y^3 + 2pxy^2 + p^2y - 1 = 0$$

$$\therefore \Delta = 0$$

$$\therefore 9p^2q^2 = -36(p+q)$$

$$\therefore (pq)^2 = -4(p+q)$$

b) eq of chord is

$$y - \frac{3}{p} = \frac{\frac{3}{p} - \frac{3}{q}}{\frac{3}{p} - \frac{3}{q}} (x - 3p)$$

Right $x = \frac{3}{2} \cdot \frac{-(pq)^2}{4} = \frac{-3}{8} (pq)^2 \quad \times$

$$y - \frac{3}{p} = \frac{-1}{pq} (x - 3p) \quad \perp$$

$$y = \frac{3}{2pq} (p+q) = \frac{3}{2pq} \frac{(pq)^2}{-4}$$

$$\therefore pqy + x = 3(p+q)$$

$$= \frac{-3}{8} (pq) \quad \times$$

mid pt of PQ is $x = \frac{3p+q}{2} = \frac{3}{2} (p+q)$

$$y = \left(\frac{\frac{3}{p} + \frac{3}{q}}{2} \right) = \frac{3(p+q)}{2pq}$$

$$\Rightarrow x = -\frac{8}{3} y^2$$

but $pq < 0 \quad \perp$

$$\therefore y > 0 \quad x < 0$$

$$c) a^x = e^{2x-1}$$

$$\ln a^x = \ln e^{2x-1} \quad |$$

$$x \ln a = (2x-1) \ln e$$

$$x \ln a = 2x - 1$$

$$2x - x \ln a = 1$$

$$x(2 - \ln a) = 1$$

$$x = \frac{1}{2 - \ln a} \quad |$$

$$a > 0 \quad \ln a \neq 2$$

(6)

6

$$a) \quad x^4 - 6x^2 + 9x^2 + 6x - 20 = 0$$

if $(2+i)$ is a zero

$2-i$ is also a zero $\perp$

$$(x - (2+i))(x - (2-i))$$

$$= x^2 - 4x + 5$$

$$(x^4 - 6x^2 + 9x^2 + 6x - 20) \div (x^2 - 4x + 5)$$

$$= x^2 - 2x - 4 \quad \perp$$

$$x^2 - 2x - 4 \Rightarrow x = \frac{2 \pm \sqrt{4+16}}{2}$$

$$= 1 \pm \sqrt{5}$$

$$= 1 \pm \sqrt{5} \quad \perp$$

$\therefore$ roots are $2 \pm i, 1 \pm \sqrt{5}$.

$$M_{AS} \cdot M_{PS}$$

$$= \frac{-b}{a} \times \frac{\frac{a^2-b^2}{b} m\theta}{a e}$$

$$= \frac{a^2-b^2}{a^2 e^2}$$

$$= \frac{a^2 - a^2 \left(\frac{e^2-1}{e^2} \right)}{a^2 e^2}$$

$$= \frac{a^2 - a^2 + a^2 e^2}{a^2 e^2}$$

$$M_{AS} \cdot M_{PS}$$

$$b) \quad \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 \quad -T$$

$$\frac{a \cos \theta}{\cos \theta} - \frac{b \sin \theta}{\sin \theta} = a^2 - b^2 - N$$

$$\text{At } \theta = 0 \text{ in } T \quad \therefore \frac{y \sin \theta}{b} = 1$$

$$y = \frac{b}{\sin \theta}$$

$$\text{At } \theta = 0 \text{ in } N \quad \left(0, \frac{b}{\sin \theta}\right)$$

$$\text{At } \theta = 0 \text{ in } N$$

$$-\frac{b \sin \theta}{\sin \theta} = a^2 - b^2$$

$$y = \frac{(b^2 - a^2) \sin \theta}{b} \quad \perp$$

$$\text{B is at } \left(0, \frac{b^2 - a^2 \sin \theta}{b}\right)$$

$$= \frac{-b}{a} \times \frac{\frac{a^2-b^2}{b} m\theta}{a e}$$

$$= \frac{-(a^2-b^2)}{a^2 e^2}$$

$$= -\frac{(a^2 - a^2 + a^2 e^2)}{a^2 e^2} \quad \perp$$

$$= -1$$

$$\therefore AS \perp PS.$$

$$\therefore \angle ASB = 90^\circ \quad \perp$$

$$\text{Now } \angle APB = 90^\circ \text{ (Tangent } \perp \text{ Normal)}$$

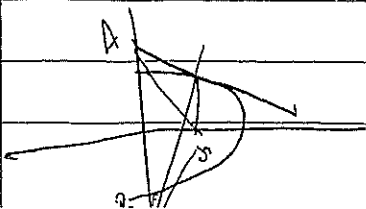
$$\angle ASD = 90^\circ$$

$\therefore$ AB subtends 90° angle at S and P $\perp$

$\therefore$ A, P, S, B are cyclic

AB is diameter.

$$\therefore \text{center is } \left(0, \left(\frac{b}{\sin \theta} + \frac{b^2 - a^2 \sin \theta}{b}\right) / 2\right)$$



C

$$\int_0^a f(x) dx$$

$$\text{let } u = a - x$$

$$du = -dx$$

$$\text{when } x=a \quad u=0$$

$$x=0 \quad u=a$$

$$\therefore I = \int_a^0 -f(a-u) du$$

$$= \int_0^a f(a-u) du$$

$$\text{let } u=x$$

$$\therefore I = \int_0^a f(a-x) dx$$

$$\int_0^\pi \frac{x \sin x}{1+\cos^2 x} dx$$

$$= \int_0^\pi \frac{(\pi-x) \sin(\pi-x)}{1+\cos^2(\pi-x)} dx$$

$$= \int_0^\pi \frac{(\pi-x) \sin x}{1+\cos^2 x} dx$$

$$= \int_0^\pi \frac{\pi \sin x - x \sin x}{1+\cos^2 x} dx$$

$$= \int_0^\pi \frac{\pi \sin x}{1+\cos^2 x} dx - \int_0^\pi \frac{x \sin x}{1+\cos^2 x} dx$$

$$I = \int_0^\pi \frac{\pi \sin x}{1+\cos^2 x} dx - I$$

$$2I = \int_0^\pi \frac{\pi \sin x}{1+\cos^2 x} dx$$

$$\text{let } u = \cos x \quad x=\pi \quad u=-1$$

$$du = -\sin x dx \quad x=0 \quad u=1$$

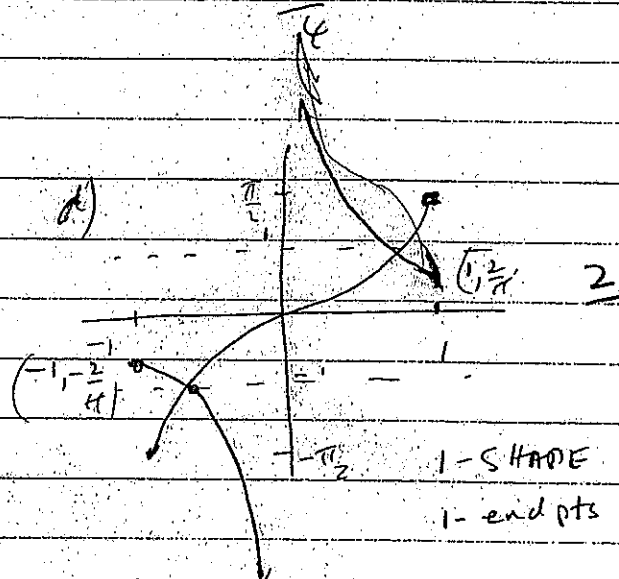
$$\therefore 2I = \pi \int_1^{-1} \frac{-du}{1+u^2}$$

$$= \left[\tan^{-1} u \right]_1^{-1}$$

$$= \left(\frac{\pi}{2} - \frac{\pi}{4} \right)$$

$$= \frac{\pi^2}{2}$$

$$I = \frac{\pi^2}{4}$$



(7)

$$a) 1) \quad 1+w+w^2=0 \quad | \cdot (1+2w+3w^2)(1+2w^2+3w) \\ = (w+2w^2)(w^2+2w) \quad | \cdot \\ = \frac{n}{2} \int (1-n)^{n-1} dx - \frac{n}{2} \int (1-2x)^n dx$$

$$= w^3 + 2w^2 + 2w^4 + 4w^3$$

$$= 1 + 2w^3 + 2w + 4$$

$$= (5 + 2(w^2 + w + 1) - 2)$$

$$= 5 - 2 = 3$$

$$= \frac{n}{2} I_{n-1} = \frac{n}{2} I_n$$

$$\therefore 1 + \frac{n}{2} I_n = \frac{n}{2} I_{n-1}$$

$$\therefore I_n = \frac{n}{n+2} I_{n-1}$$

$$I_1 = \frac{1}{3} I_0$$

$$\text{but } I_0 = \int_0^1 dx = [x]_0^1 = 1$$

$$\therefore I_1 = \frac{1}{3}$$

$$I_2 = \frac{2}{5} \cdot \frac{1}{3}$$

$$I_3 = \frac{3}{5} \cdot \frac{2}{5} \cdot \frac{1}{3}$$

1) Given root with the

$$1, w \text{ or } w^2$$

$$\text{if } x=1 \quad p+q+r=0$$

$$x=w \quad pw^5+qw+r=0$$

$$x=w^2 \quad pw^{10}+qw^2+r=0$$

$$\therefore (p+q+r)/(pw^5+qw+r)/(pw^{10}+qw^2+r) = 26$$

$$b) 1) \quad (1-\sqrt{x})^{n-1} - (1-\sqrt{x})^n$$

$$= (1-\sqrt{x})^{n-1} (1 - (1-\sqrt{x}))$$

$$= (1-\sqrt{x})^{n-1} (1 - 1 + \sqrt{x})$$

$$= \sqrt{x} (1-\sqrt{x})^{n-1}$$

$$\therefore I_{100} = \frac{100 \cdot 99 \cdot 98 \cdot 97 \dots 3 \cdot 2 \cdot 1}{102 \cdot 101 \cdot 100 \cdot 99 \dots 5 \cdot 4 \cdot 3}$$

$$= \frac{2}{102 \cdot 101}$$

$$11) \quad I_n = \int_0^1 (1-\sqrt{x})^n dx$$

$$\text{Let } u = (1-\sqrt{x})^n \quad v' = 1$$

$$u' = n(1-\sqrt{x})^{n-1} \cdot \frac{-1}{2\sqrt{x}} \quad v = x$$

$$\therefore I_n = \left[\frac{2(1-\sqrt{x})^n}{n} \right]_0^1 + \frac{n}{2} \int \frac{(1-\sqrt{x})^{n-1}}{\sqrt{x}} \cdot x dx$$

$$= 0 + \frac{n}{2} \int (1-\sqrt{x})^{n-1} \sqrt{x} dx$$

7C)

$$V = \frac{1}{3} \pi r^2 h$$

$$qV^2 = \pi^2 r^4 h^2$$

new

$$S = \pi r l$$

$$\text{when } l^2 = h^2 + r^2$$

$$\therefore S = \pi r \sqrt{h^2 + r^2} \quad \perp$$

$$h^2 + r^2 = \frac{S^2}{\pi^2 r^2}$$

$$h^2 = \frac{S^2}{\pi^2 r^2} - r^2$$

$$h^2 = \frac{S^2 - \pi^2 r^4}{\pi^2 r^2}$$

$$\therefore qV^2 = \pi^2 r^4 \left(\frac{S^2 - \pi^2 r^4}{\pi^2 r^2} \right)$$

$$qV^2 = r^2 (S^2 - \pi^2 r^4) \quad \perp$$

new Volume when qV^2 is max

$$\therefore \frac{d qV^2}{dr} = 2rS^2 - 6\pi^2 r^5$$

$$= 0 \text{ when}$$

$$r=0 \text{ or } r^4 = \frac{S^2}{3\pi^2} \quad \perp$$

$$\text{or } r^2 = \frac{S}{3^{1/2} \pi}$$

new

$$V = \frac{1}{3} \pi r^2 h \quad \perp$$

$$= \frac{1}{3} \pi \cdot \frac{S}{3^{1/2} \pi} \cdot \sqrt{\frac{S^2 - \pi^2 r^4}{\pi^2 r^2}}$$

$$= \frac{S}{3^{3/2}} \sqrt{\frac{S^2 - \frac{\pi^2 S^2}{3\pi^2}}{\pi^2 \cdot \frac{S}{3^{1/2} \pi}}}$$

$$= \frac{S}{3^{3/2}} \sqrt{\frac{\frac{2}{3} S^2 \cdot \frac{3^{1/2}}{S\pi}}{1}}$$

$$= \frac{S}{3^{3/2}} \sqrt{\frac{2S}{3^{1/2} \pi}}$$

$$= \frac{S}{3^{3/2}} \cdot \frac{2^{1/2} S^{1/2}}{3^{1/4} \pi^{1/2}}$$

$$= \frac{2^{1/2} S^{3/2}}{3^{7/4} \pi^{1/2}} \quad \perp$$

8a) when $n=1$

$$7^n + 3n(7^n) - 1$$

$$= 7 + 3 \cdot 7 - 1$$

$$= 7 + 21 - 1 = 27$$

which is $\div$ by 9

$\therefore$ true for $n=1$

same true for some value k

Let $7^k + 3k(7^k) - 1 = 9N$

where N is an integer

not prove $7^{k+1} + 3(k+1)(7^{k+1}) - 1$

is $\div$ by 9

$$= 7 \cdot 7^k + 3k \cdot 7 \cdot 7^k + 3 \cdot 7 \cdot 7^k - 1$$

$$= 7(7^k + 3k \cdot 7^k - 1) + 7 + 21 \cdot 7^k - 1$$

$$= 7 \cdot 9N + 6 + 21 \cdot 7^k$$

$$= 9(7N) + 3(2 + 7 \cdot 7^k)$$

which is $\div$ by 9 if $2 + 7 \cdot 7^k$ is $\div$ by 3

if $k=1$ $2 + 7 \cdot 7^k = 2 + 49 = 51 \div$ by 3

assume $2 + 7^{k+1} = 9P$ (P is an integer)

not prove $2 + 7^{k+2}$ is $\div$ by 3

$$2 + 7^{k+2} = 2 + 7 \cdot 7^{k+1}$$

$$= 7(2 + 7 \cdot 7^{k+1}) - 12$$

$$= 21P - 12$$

$$= 3(7P - 4) \text{ which is } \div \text{ by } 3$$

$$7^{k+1} + 3(k+1)7^{k+1} - 1 \text{ is } \div \text{ by } 9$$

$\therefore$ as true for $n=1$ and true for $n=k$

but for $n=2$ $n=3$

and so on for all n

$$7^n + 3n(7^n) - 1 \text{ is } \div \text{ by } 9.$$

b) $k_{40} = 1$

$$40 = \frac{\pi}{4} + 2k\pi \quad k=0, 71, \dots$$

$$\text{or } \frac{5\pi}{4} + 2k\pi \quad k=0, 71, \dots$$

$$\text{or } 40 = \frac{\pi}{4} + k\pi \quad k=0, 71, \dots$$

$$\therefore 0 = \frac{\pi}{4} + \frac{k}{4}\pi \quad k=0, 71, \dots$$

$$(\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta$$

$$\therefore \cos 4\theta + i \sin 4\theta = \cos^4 \theta + 4i \cos^3 \theta \sin \theta + 6i^2 \cos^2 \theta \sin^2 \theta + 4i^3 \cos \theta \sin^3 \theta + i^4 \sin^4 \theta$$

$$= (\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta) + i(4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta)$$

$$\therefore \cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

$$\sin 4\theta = 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta$$

$$\frac{\sin 4\theta}{\cos 4\theta} = \frac{4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta}{\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta}$$

$\div$ top & bottom by $\cos^4 \theta$

$$\therefore \tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 4 \tan^2 \theta + \tan^4 \theta}$$

$$\text{Let } \tan 4\theta = 1 \text{ and } \tan \theta = x$$

$$\therefore \frac{4x - 4x^3}{1 - 4x^2 + x^4} = 1$$

$$x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$$

$$\text{when } x = \tan \theta$$

$$\therefore x = \tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, \tan \frac{9\pi}{16}, \tan \frac{13\pi}{16}$$

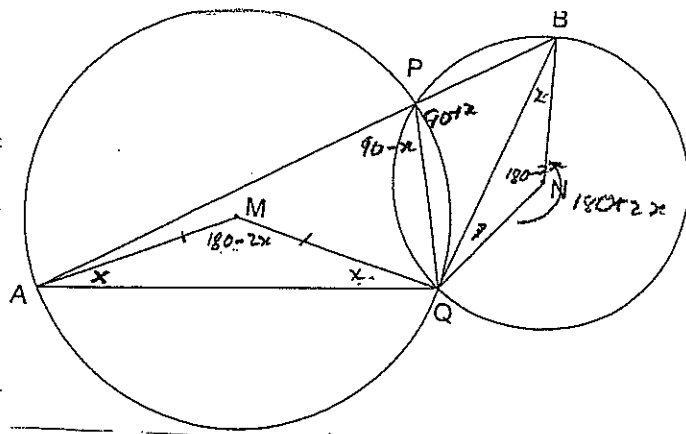
$$\sum x^2 = (\sum x)^2 - 2 \sum x_1 x_2$$

$$= (-4)^2 - 6 \cdot -2 = 16 + 12 = 28$$

$$\tan^2 \frac{\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{9\pi}{16} + \tan^2 \frac{13\pi}{16} = 28$$

$$\text{but } \tan^2 \frac{9\pi}{16} = \tan^2 \frac{7\pi}{16} \text{ and } \tan^2 \frac{13\pi}{16} = \tan^2 \frac{3\pi}{16}$$

$$\therefore \tan^2 \frac{\pi}{16} + \tan^2 \frac{3\pi}{16} + \tan^2 \frac{7\pi}{16} + \tan^2 \frac{5\pi}{16} = 28$$



let $\angle MAQ = x^\circ$

$AM = MQ$ (equal radii)

$\therefore \triangle AMQ$ is isosceles

$\therefore \angle MAQ = \angle MQA = x$

(1)

$\angle AMQ = 180 - 2\angle MAQ$ (angle sum of triangle)

$\angle AMQ = 180 - 2x$

$\angle APQ = \frac{1}{2} \angle AMQ$ (angle at centre is twice angle at circumference) (1)

$\angle APQ = 90 - x$

$\angle APQ + \angle BPQ = 180^\circ$ (str line)

$\therefore \angle BPQ = 90 + x$

reflex $\angle BNQ = 2\angle BPQ$ (" ") (1)

OBTUSE $\angle BNQ = 180 + 2x$

$\therefore \angle BNQ = 180 - 2x$

but $\angle NBA = \angle NAB$ ($NB = NA$ equal radii)

$\angle NBA = 180 - (180 - 2x)$

$= x$

$= \angle MAQ$

(1)

4

$\therefore \angle NBA = \angle MAQ$